# case_design

April 25, 2026

### 0.0.1 Case design analysis

Experimental case selection and protocol design for human-AI decision support system study

- Design: 18 cases × 3 protocols (within-subject)
- Each participant: 18 trials (~20 min)
- Target N: 100 participants (~33 per protocol group)

[1]:
```
import os
import sys
import json
import warnings
warnings.filterwarnings('ignore')

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from pathlib import Path

ROOT       = Path.cwd().parents[1]          # capstone/
ARTIFACTS  = ROOT / "artifacts"
BUILD_DIR  = ARTIFACTS / "build"
FROZEN_DIR = ARTIFACTS / "frozen"
FIGS_DIR   = ROOT / "codes" / "visualizations"
BUILD_DIR.mkdir(parents=True, exist_ok=True)
FIGS_DIR.mkdir(parents=True, exist_ok=True)

sys.path.append(str(ROOT / "codes"))
from feature_config import FEATURES, TARGET

np.random.seed(42)
pd.set_option('display.max_columns', None)
sns.set_style("whitegrid")

print("ROOT       :", ROOT)
print("BUILD_DIR  :", BUILD_DIR)
print("FROZEN_DIR :", FROZEN_DIR)
print("FIGS_DIR   :", FIGS_DIR)
```

```
ROOT       : /Users/annaasatryan/Desktop/capstone
BUILD_DIR  : /Users/annaasatryan/Desktop/capstone/artifacts/build
FROZEN_DIR : /Users/annaasatryan/Desktop/capstone/artifacts/frozen
FIGS_DIR   : /Users/annaasatryan/Desktop/capstone/codes/visualizations
```

[2]:
```python
# ================================================================
# FREEZE GUARD
# Re-running this notebook writes regenerated working outputs to
# artifacts/build/ only. artifacts/frozen/ is the official locked
# experiment snapshot and is NEVER overwritten here.
# Freezing happens later via an explicit freeze step only.
# Run `python run.py --mode validate` to verify frozen integrity.
# ================================================================
_lock = FROZEN_DIR / "cases.lock.json"
if _lock.exists():
    _locked = json.loads(_lock.read_text())
    print("=" * 62)
    print("FROZEN ARTIFACTS LOCKED")
    print(f"  git commit : {str(_locked.get('git_commit', 'unknown'))[:12]}")
    print(f"  locked at  : {_locked.get('created_at', 'unknown')}")
    print()
    print("  Re-running writes to artifacts/build/ (working copies).")
    print("  artifacts/frozen/ is protected - run validate to confirm.")
    print("=" * 62)
else:
    print("No frozen artifacts - fresh case selection run.")
```

```
==============================================================
FROZEN ARTIFACTS LOCKED
  git commit : d77f40aeca21
  locked at  : 2026-04-22T23:21:54.001700+00:00

  Re-running writes to artifacts/build/ (working copies).
  artifacts/frozen/ is protected - run validate to confirm.
==============================================================
```

[3]:
```python
# Load from build candidate pool (generated by modeling.ipynb).
# Once frozen, the freeze step locks candidate_pool_scored.parquet in
# artifacts/frozen/. This notebook always reads from BUILD_DIR so it
# can be re-run independently of the freeze state.
pool_path = BUILD_DIR / "candidate_pool_scored.parquet"
df = pd.read_parquet(pool_path)

df['pred_prob'] = df['pred_prob'].clip(1e-6, 1 - 1e-6)

print(f"Loaded: {pool_path}")
print("Shape:", df.shape)
print("Default rate:", df['y_true'].mean().round(4))
```

```
print("Difficulty tier counts:")
print(df['difficulty_tier'].value_counts())
```

```
Loaded: /Users/annaasatryan/Desktop/capstone/artifacts/build/candidate_pool_scor
ed.parquet
Shape: (479744, 14)
Default rate: 0.2288
Difficulty tier counts:
difficulty_tier
easy      241081
medium    175439
hard       63224
Name: count, dtype: int64
```

**Feature engineering**

[4]:
```
# correctness: model correct at 0.5 threshold
#   used ONLY for case stratification, not DSS decision-making
#
# confidence: probability assigned to the predicted class
#   measures how certain the model is, regardless of direction

df['model_pred_class'] = (df['pred_prob'] >= 0.5).astype(int)
df['correct']          = (df['model_pred_class'] == df['y_true']).astype(int)

df['confidence'] = np.where(
    df['pred_prob'] >= 0.5,
    df['pred_prob'],
    1 - df['pred_prob']
)

# Confidence bins within difficulty tier
# Avoids confounding between difficulty and confidence
df['conf_bin'] = (
    df.groupby('difficulty_tier', observed=True)['confidence']
    .transform(lambda x: pd.qcut(x, q=2, labels=['low', 'high']))
)
```

**Validation checks**

[5]:
```
print("=== ACCURACY BY DIFFICULTY ===")
print(df.groupby('difficulty_tier', observed=True)['correct'].mean().round(4))

print("\n=== CONFIDENCE BY DIFFICULTY ===")
print(df.groupby('difficulty_tier', observed=True)['confidence'].mean().
  ↪round(4))
```

```
=== ACCURACY BY DIFFICULTY ===
difficulty_tier
```

```
easy      0.8629
hard      0.5888
medium    0.7014
Name: correct, dtype: float64

=== CONFIDENCE BY DIFFICULTY ===
difficulty_tier
easy      0.8877
hard      0.6284
medium    0.7334
Name: confidence, dtype: float64
```

**Confidence-accuracy alignment diagnostic**

[6]:
```
gap = (
    df[df['correct'] == 1]['confidence'].mean()
    - df[df['correct'] == 0]['confidence'].mean()
)

print(f"Mean confidence when correct   : "
      f"{df[df['correct']==1]['confidence'].mean():.4f}")
print(f"Mean confidence when incorrect : "
      f"{df[df['correct']==0]['confidence'].mean():.4f}")
print(f"Confidence gap (correct - wrong): {gap:.4f}")
```

```
Mean confidence when correct   : 0.8170
Mean confidence when incorrect : 0.7313
Confidence gap (correct - wrong): 0.0857
```

Gap of ~0.086 indicates moderate confidence-accuracy alignment. The model is more confident when correct than when wrong, but the separation is not large enough to make reliance decisions trivial. This creates a non-degenerate environment for studying human reliance behavior. Directly relevant to H5 (confidence alignment hypothesis).

**Directional error analysis**

[7]:
```
df['fp'] = ((df['model_pred_class'] == 1) & (df['y_true'] == 0)).astype(int)
df['fn'] = ((df['model_pred_class'] == 0) & (df['y_true'] == 1)).astype(int)

print("=== Error types by difficulty ===")
print(df.groupby('difficulty_tier', observed=True)[['fp','fn']].mean().round(4))
```

```
=== Error types by difficulty ===
                     fp      fn
difficulty_tier
easy             0.0000  0.1371
hard             0.1898  0.2214
medium           0.0434  0.2552
```

FP = model predicts default, applicant actually paid (false alarm) FN = model predicts paid, applicant actually defaulted (missed default) FN dominates in medium/easy — model rarely predicts

default at 0.5 threshold. This asymmetry is WHY the DSS uses probabilities, not binary decisions.

**Normative decision benchmark**

- C_FN = cost of approving a defaulting loan (missed default)
- C_FP = cost of rejecting a creditworthy loan (missed opportunity)

C_FN:C_FP = 5:1 is a **design assumption**, consistent with standard credit risk literature where losses from defaults typically exceed foregone interest income by a ratio of 4–6×. This yields tau = 1/6  0.167.

This ratio is used for normative diagnostics only (model-optimal agreement, cost benchmarks). It is a fixed parameter of the experimental design, not a data-derived estimate.

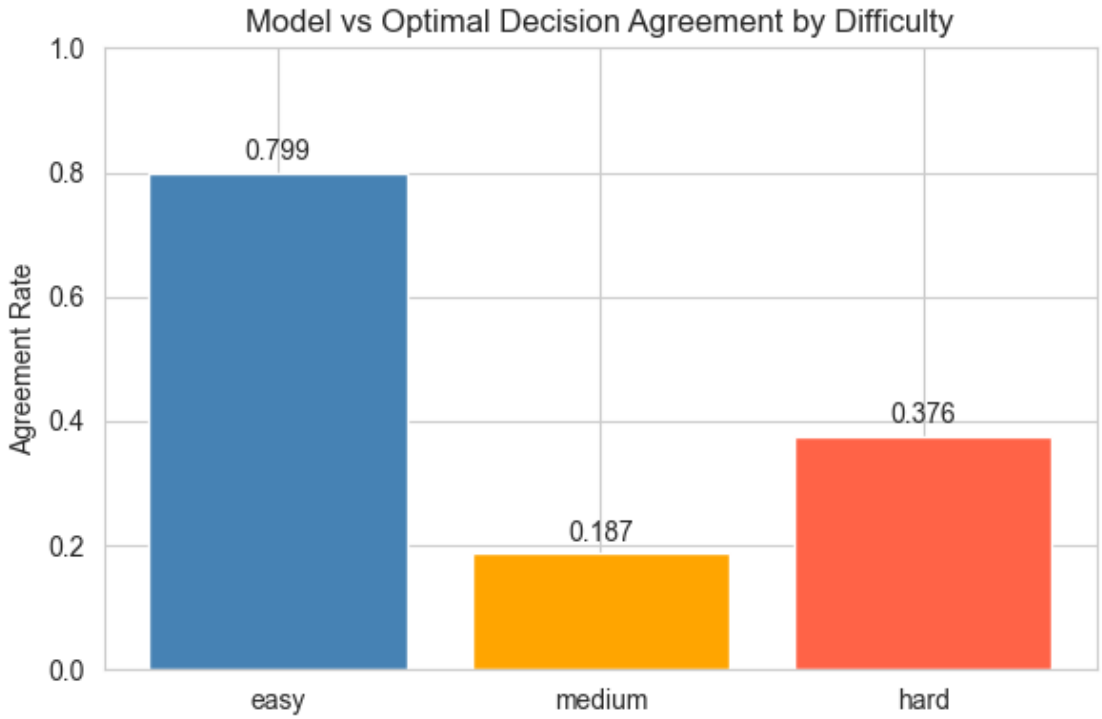
[8]:
```
C_FN = 5
C_FP = 1
tau  = C_FP / (C_FP + C_FN)

df['model_decision']   = (df['pred_prob'] < 0.5).astype(int)   # 1=approve
df['optimal_decision'] = (df['pred_prob'] < tau).astype(int)   # 1=approve
df['model_optimal']    = (df['model_decision'] == df['optimal_decision']).
 ↪astype(int)

print(f"Decision threshold tau: {tau:.4f}")
print(f"\nApproval rate - model (0.5):    {df['model_decision'].mean():.4f}")
print(f"Approval rate - optimal (tau):  {df['optimal_decision'].mean():.4f}")
print(f"Model-optimal disagreement:     "
      f"{(df['model_decision'] != df['optimal_decision']).mean():.4f}")

print("\n=== Model-optimal agreement by difficulty ===")
print(df.groupby('difficulty_tier', observed=True)['model_optimal'].mean().
 ↪round(4))
```

```
Decision threshold tau: 0.1667

Approval rate - model (0.5):    0.9217
Approval rate - optimal (tau):  0.4408
Model-optimal disagreement:     0.4809

=== Model-optimal agreement by difficulty ===
difficulty_tier
easy      0.7987
hard      0.3758
medium    0.1865
Name: model_optimal, dtype: float64
```

Medium-difficulty cases show the LOWEST model-optimal agreement (18.7%), lower than even hard cases (37.6%). This is expected: medium-difficulty cases have base rates closest to the cost-sensitive threshold tau=0.167, where the standard 0.5-threshold model most frequently disagrees with the optimal cost-sensitive rule. This is not an anomaly — it is a feature of the experimental

design that makes medium cases the most decision-theoretically consequential.

[9]:
```
fig, ax = plt.subplots(figsize=(6, 4))
order = ['easy', 'medium', 'hard']
vals  = [
    df[df['difficulty_tier'] == t]['model_optimal'].mean()
    for t in order
]
ax.bar(order, vals, color=['steelblue', 'orange', 'tomato'])
ax.set_title("Model vs Optimal Decision Agreement by Difficulty")
ax.set_ylabel("Agreement Rate")
ax.set_ylim(0, 1)
for i, v in enumerate(vals):
    ax.text(i, v + 0.02, f"{v:.3f}", ha='center', fontsize=10)
plt.tight_layout()
plt.savefig(FIGS_DIR / "normative_agreement.png", dpi=150, bbox_inches='tight')
plt.show()
```




## EXPERIMENTAL DESIGN

1. Each participant completes exactly **18 trials** (one per case). Across the 3 participant groups, each case is observed under every protocol.
2. Protocol assignment is counterbalanced:
    - Group 1 (n 33): Cases 1–6 → no_ai | 7–12 → human_first | 13–18 → ai_first

- Group 2 (n 33): Cases 1–6 → human_first | 7–12 → ai_first | 13–18 → no_ai
- Group 3 (n 34): Cases 1–6 → ai_first | 7–12 → no_ai | 13–18 → human_first

This ensures every case is seen under every protocol across the full sample. With N=100: ~33 participants per protocol condition per case cluster.

3. CASE STRATIFICATION: difficulty (3) × model_correctness (2) = 6 cells × 3 cases = 18 cases

   **WHY THESE CELLS:**

   - easy+correct → baseline, participant should agree with AI
   - easy+wrong → automation bias check (AI is confident but wrong)
   - medium+correct → moderate uncertainty, correct AI
   - medium+wrong → medium uncertainty, wrong AI — normative gap is largest here
   - hard+correct → under-reliance check (AI uncertain but correct)
   - hard+wrong → highest ambiguity, wrong AI

4. Confidence is recorded as a covariate, not a design factor. This preserves statistical power while maintaining interpretability.

**Stratified case selection**

[10]:
```
N_PER_CELL = 3

# --- Filter pool ---
df_pool = df[
    (df['pred_prob'] > 0.05) &
    (df['pred_prob'] < 0.95) &
    (df['difficulty_tier'].notna()) &
    (df['purpose'] != 'other')
].copy()

# Ordered categorical so groupby / sort respects logical order
_tier_order = pd.CategoricalDtype(['easy', 'medium', 'hard'], ordered=True)
df_pool['difficulty_tier'] = df_pool['difficulty_tier'].astype(_tier_order)

print(f"Case pool after filtering: {len(df_pool):,}")
print(df_pool.groupby(['difficulty_tier', 'correct'], observed=True).size().
 ↪unstack(fill_value=0))


def sample_cell(df, tier, correct, n, already_selected_probs=None, min_dist=0.
 ↪02):
    subset = df[
        (df['difficulty_tier'] == tier) &
        (df['correct'] == correct)
    ].copy()

    if len(subset) <= n:
```

```
        return subset

    if already_selected_probs is None:
        already_selected_probs = []

    if tier == 'hard':
        subset = subset.sort_values('pred_prob', ascending=False)
    elif tier == 'easy' and correct == 1:
        subset = subset.sort_values('pred_prob', ascending=True)
    elif 'model_optimal' in subset.columns and subset['model_optimal'].
↪nunique() > 1:
        subset = subset.sort_values(['model_optimal', 'pred_prob'])
    else:
        subset = subset.sort_values('pred_prob')

    selected_rows  = []
    selected_probs = list(already_selected_probs)
    used_purposes  = set()

    # Pass 1: enforce distance AND purpose diversity
    for _, row in subset.iterrows():
        prob = row['pred_prob']
        if any(abs(prob - p) < min_dist for p in selected_probs):
            continue
        if row['purpose'] not in used_purposes:
            selected_rows.append(row)
            selected_probs.append(prob)
            used_purposes.add(row['purpose'])
        if len(selected_rows) >= n:
            break

    # Pass 2: relax purpose diversity, keep distance
    if len(selected_rows) < n:
        remaining = subset.drop(index=[r.name for r in selected_rows])
        for _, row in remaining.iterrows():
            prob = row['pred_prob']
            if any(abs(prob - p) < min_dist for p in selected_probs):
                continue
            selected_rows.append(row)
            selected_probs.append(prob)
            if len(selected_rows) >= n:
                break

    # Pass 3: relax spacing - deterministic stable tie-breaker
    if len(selected_rows) < n:
        remaining = (
            subset
```

```
            .drop(index=[r.name for r in selected_rows])
            .sort_values(['pred_prob', 'case_id'])
        )
        for _, row in remaining.iterrows():
            selected_rows.append(row)
            if len(selected_rows) >= n:
                break

    return pd.DataFrame(selected_rows)


# Order matters: easy-correct sampled FIRST to protect low-probability cases
# before cross-cell distance constraint blocks them
cells = [
    ('easy', 1),
    ('hard', 1),
    ('hard', 0),
    ('medium', 1),
    ('medium', 0),
    ('easy', 0),
]

all_selected_probs = []
sampled = []

for tier, correct in cells:
    cell_df = sample_cell(
        df_pool,
        tier,
        correct,
        N_PER_CELL,
        already_selected_probs=all_selected_probs,
        min_dist=0.02
    )
    all_selected_probs.extend(cell_df['pred_prob'].tolist())
    sampled.append(cell_df)

cases = (
    pd.concat(sampled)
    .drop_duplicates()
    .reset_index(drop=True)
)
cases['case_position'] = cases.index + 1

print(f"\nSELECTED: {len(cases)}")
```

```
Case pool after filtering: 417,304
correct              0       1
```

```
difficulty_tier
easy              29621  168249
medium            48155  112787
hard              24215   34277

SELECTED: 18
```

[11]:
```python
C_FN = 5  # cost of approving a defaulter
C_FP = 1  # cost of rejecting a good loan

def compute_cost(decisions, y_true, c_fn=5, c_fp=1):
    """decisions: 1=approve, 0=reject"""
    fn = ((decisions == 1) & (y_true == 1)).sum() * c_fn
    fp = ((decisions == 0) & (y_true == 0)).sum() * c_fp
    return (fn + fp) / len(y_true)

y = cases['y_true'].values

# Strategy baselines
always_approve = np.ones(18)
always_reject = np.zeros(18)
model_05 = (cases['pred_prob'] < 0.5).astype(int).values  # approve if pred < 0.
↪5
model_tau = (cases['pred_prob'] < tau).astype(int).values   # approve if pred <␣
↪tau
random_decision = np.random.binomial(1, 0.5, 18)

print("=== EXPECTED COST PER CASE (18-case set) ===\n")
print(f"Always approve : {compute_cost(always_approve, y):.3f}")
print(f"Always reject  : {compute_cost(always_reject, y):.3f}")
print(f"Model at 0.5   : {compute_cost(model_05, y):.3f}")
print(f"Optimal at tau : {compute_cost(model_tau, y):.3f}")
print(f"Random          : {compute_cost(random_decision, y):.3f}")
```

```
=== EXPECTED COST PER CASE (18-case set) ===

Always approve : 2.500
Always reject  : 0.500
Model at 0.5   : 1.833
Optimal at tau : 0.333
Random          : 1.000
```

### Protocol assignment

[12]:
```python
np.random.seed(42)

# Convert to ordered categorical before block assignment so sorting is␣
↪logically ordered
```

```
_tier_order = pd.CategoricalDtype(['easy', 'medium', 'hard'], ordered=True)
cases['difficulty_tier'] = cases['difficulty_tier'].astype(_tier_order)

block_labels = ['block_1', 'block_2', 'block_3']
assignments  = []

for (diff, corr), group in cases.groupby(['difficulty_tier', 'correct'],␣
 ↪observed=True):
    group_shuffled = group.sample(frac=1, random_state=42).
 ↪reset_index(drop=True)

    assert len(group_shuffled) == 3, (
        f"Expected 3 cases for ({diff}, correct={corr}), got␣
 ↪{len(group_shuffled)}"
    )

    for i, block in enumerate(block_labels):
        row = group_shuffled.iloc[i].copy()
        row['block'] = block
        assignments.append(row)

cases = pd.DataFrame(assignments).reset_index(drop=True)
cases['difficulty_tier'] = cases['difficulty_tier'].astype(_tier_order)

cases = cases.sort_values(['block', 'difficulty_tier', 'correct']).
 ↪reset_index(drop=True)
cases['case_position'] = cases.index + 1

protocol_rotation = pd.DataFrame({
    'participant_group': ['group_1', 'group_2', 'group_3'],
    'block_1_protocol' : ['no_ai',       'human_first', 'ai_first'],
    'block_2_protocol' : ['human_first', 'ai_first',    'no_ai'],
    'block_3_protocol' : ['ai_first',    'no_ai',       'human_first'],
    'target_n'         : [33, 33, 34]
})

print("=== PROTOCOL ROTATION TABLE ===\n")
print(protocol_rotation.to_string(index=False))

#   Hard protocol checks
assert len(protocol_rotation) == 3, "Must have exactly 3 participant groups"
assert set(protocol_rotation['participant_group']) == {'group_1', 'group_2',␣
 ↪'group_3'}, \
    "Participant groups must be group_1, group_2, group_3"

_expected_protocols = {'no_ai', 'human_first', 'ai_first'}
for _col in ['block_1_protocol', 'block_2_protocol', 'block_3_protocol']:
```

```
    assert set(protocol_rotation[_col]) == _expected_protocols, \
        f"{_col} must contain exactly {_expected_protocols}"

for _, _row in protocol_rotation.iterrows():
    _group_protocols = {
        _row['block_1_protocol'],
        _row['block_2_protocol'],
        _row['block_3_protocol']
    }
    assert _group_protocols == _expected_protocols, \
        f"Group {_row['participant_group']} does not have all 3 protocols"

assert cases['block'].nunique() == 3, "Must have exactly 3 blocks"
assert (cases.groupby('block', observed=True).size() == 6).all(), \
    "Each block must have exactly 6 cases"

for _block in block_labels:
    _sub = cases[cases['block'] == _block]
    _ct  = pd.crosstab(_sub['difficulty_tier'], _sub['correct'])
    assert (_ct == 1).all().all(), \
        f"Block {_block} must have exactly 1 case per difficulty × correctness␣
 ↪cell"

print("\n  All protocol checks passed")
print("\n\n=== DESIGN VERIFICATION ===\n")
```

```
=== PROTOCOL ROTATION TABLE ===

participant_group block_1_protocol block_2_protocol block_3_protocol  target_n
          group_1            no_ai      human_first         ai_first        33
          group_2      human_first         ai_first            no_ai        33
          group_3         ai_first            no_ai      human_first        34

  All protocol checks passed


=== DESIGN VERIFICATION ===
```

[13]:
```
# Check 1: Block × Difficulty must be fully crossed
ct_block_diff = pd.crosstab(cases['block'], cases['difficulty_tier'])
print("Block × Difficulty (should be 2 per cell):")
print(ct_block_diff)
assert (ct_block_diff == 2).all().all(), "FAIL: Block × Difficulty not balanced!
 ↪"

# Check 2: Block × Correctness must be balanced
```

```
ct_block_corr = pd.crosstab(cases['block'], cases['correct'])
print("\nBlock × Correctness (should be 3 per cell):")
print(ct_block_corr)
assert (ct_block_corr == 3).all().all(), "FAIL: Block × Correctness not␣
 ↪balanced!"

# Check 3: Within each block, difficulty × correctness = 1 per cell
for block in block_labels:
    sub = cases[cases['block'] == block]
    ct = pd.crosstab(sub['difficulty_tier'], sub['correct'])
    assert (ct == 1).all().all(), f"FAIL: {block} not fully crossed!"

print("\nWithin-block Difficulty × Correctness:  (1 per cell in every block)")

# Check 4: Total counts
print(f"\nTotal cases: {len(cases)}")
print(f"Cases per block: {cases['block'].value_counts().to_dict()}")
print(f"y_true balance: {cases['y_true'].value_counts().to_dict()}")

print("\n=== ALL CHECKS PASSED ===")
print("\nEach participant group now sees EVERY difficulty under their assigned␣
 ↪protocol.")
print("The protocol × difficulty interaction (H2) is identifiable␣
 ↪within-subject.")
```

```
Block × Difficulty (should be 2 per cell):
difficulty_tier  easy  medium  hard
block
block_1             2       2     2
block_2             2       2     2
block_3             2       2     2

Block × Correctness (should be 3 per cell):
correct  0  1
block
block_1  3  3
block_2  3  3
block_3  3  3

Within-block Difficulty × Correctness:  (1 per cell in every block)

Total cases: 18
Cases per block: {'block_1': 6, 'block_2': 6, 'block_3': 6}
y_true balance: {1: 9, 0: 9}

=== ALL CHECKS PASSED ===
```

Each participant group now sees EVERY difficulty under their assigned protocol.
The protocol × difficulty interaction (H2) is identifiable within-subject.

### Distribution checks

[14]:
```
print("=== CASE DISTRIBUTION ===\n")

print("By difficulty:")
print(cases['difficulty_tier'].value_counts().sort_index())

print("\nBy correctness:")
print(cases['correct'].value_counts())

print("\nBy block:")
print(cases['block'].value_counts().sort_index())

print("\nCross-tab (difficulty × correctness):")
print(pd.crosstab(cases['difficulty_tier'], cases['correct']))

print("\nPredicted probability by cell:")
print(
    cases.groupby(
        ['difficulty_tier', 'correct'], observed=True
    )['pred_prob'].agg(['min', 'mean', 'max']).round(3)
)

print("\nConfidence by cell:")
print(
    cases.groupby(
        ['difficulty_tier', 'correct'], observed=True
    )['confidence'].agg(['min', 'mean', 'max']).round(3)
)
```

```
=== CASE DISTRIBUTION ===

By difficulty:
difficulty_tier
easy      6
medium    6
hard      6
Name: count, dtype: int64

By correctness:
correct
0    9
1    9
Name: count, dtype: int64

By block:
```

```
block
block_1    6
block_2    6
block_3    6
Name: count, dtype: int64

Cross-tab (difficulty × correctness):
correct          0  1
difficulty_tier
easy             3  3
medium           3  3
hard             3  3

Predicted probability by cell:
                           min   mean    max
difficulty_tier correct
easy            0        0.333  0.367  0.397
                1        0.055  0.077  0.102
medium          0        0.234  0.266  0.296
                1        0.169  0.189  0.210
hard            0        0.779  0.836  0.900
                1        0.858  0.909  0.946

Confidence by cell:
                           min   mean    max
difficulty_tier correct
easy            0        0.603  0.633  0.667
                1        0.898  0.923  0.945
medium          0        0.704  0.734  0.766
                1        0.790  0.811  0.831
hard            0        0.779  0.836  0.900
                1        0.858  0.909  0.946
```

### Probability distribution

[15]:
```
print("\n=== PROBABILITY SUMMARY ===\n")
print(cases['pred_prob'].describe())
```

```
=== PROBABILITY SUMMARY ===

count    18.000000
mean      0.440742
std       0.328978
min       0.054726
25%       0.194111
50%       0.314604
75%       0.818032
max       0.946130
```

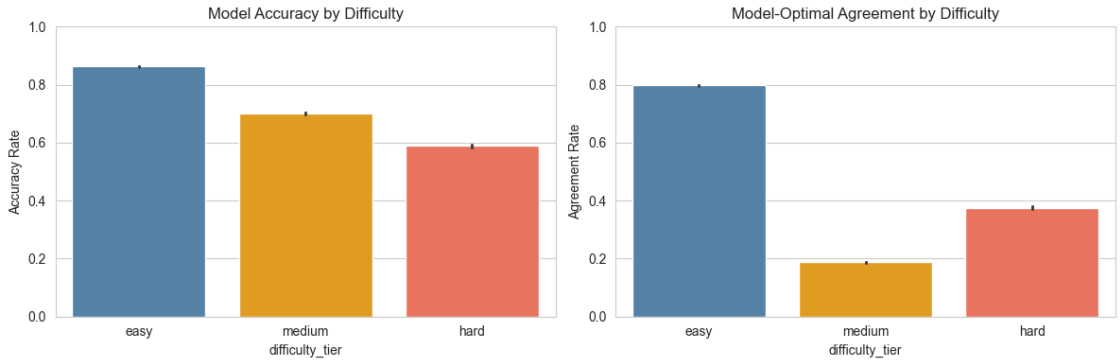
```
Name: pred_prob, dtype: float64
```

**Difficulty x correctness interaction**

[16]:
```
print("=== DIFFICULTY × CORRECTNESS INTERACTION (full test set) ===\n")
interaction = (
    df.groupby(['difficulty_tier', 'correct'], observed=True)
    .agg(
        n          =('pred_prob', 'size'),
        avg_pred   =('pred_prob', 'mean'),
        avg_conf   =('confidence','mean'),
        model_opt  =('model_optimal','mean')
    )
    .round(3)
)
print(interaction.to_string())

fig, axes = plt.subplots(1, 2, figsize=(12, 4))

sns.barplot(
    data=df, x='difficulty_tier', y='correct',
    hue='difficulty_tier', palette=['steelblue','orange','tomato'],
    ax=axes[0], legend=False, order=['easy','medium','hard']
)
axes[0].set_title("Model Accuracy by Difficulty")
axes[0].set_ylabel("Accuracy Rate")
axes[0].set_ylim(0, 1)

sns.barplot(
    data=df, x='difficulty_tier', y='model_optimal',
    hue='difficulty_tier', palette=['steelblue','orange','tomato'],
    ax=axes[1], legend=False, order=['easy','medium','hard']
)
axes[1].set_title("Model-Optimal Agreement by Difficulty")
axes[1].set_ylabel("Agreement Rate")
axes[1].set_ylim(0, 1)

plt.tight_layout()
plt.savefig(FIGS_DIR / "difficulty_interactions.png", dpi=150,␣
 ↪bbox_inches='tight')
plt.show()
```

```
=== DIFFICULTY × CORRECTNESS INTERACTION (full test set) ===

                           n  avg_pred  avg_conf  model_opt
difficulty_tier correct
easy            0      33044     0.137     0.863      0.672
                1     208037     0.108     0.892      0.819
```

```
hard            0         26000     0.459     0.615     0.462
                1         37224     0.419     0.638     0.316
medium          0         52384     0.312     0.706     0.210
                1        123055     0.262     0.745     0.177
```




### Final export

[17]:
```python
EXPORT_COLS = [
    'case_id', 'case_position', 'block',
    'loan_amnt', 'term', 'int_rate', 'log_annual_inc',
    'dti', 'revol_util', 'home_ownership', 'purpose',
    'credit_history_years',
    'pred_prob', 'y_true', 'difficulty_tier', 'difficulty_score',
    'correct', 'confidence', 'conf_bin', 'model_optimal'
]

final_cases = cases[[c for c in EXPORT_COLS if c in cases.columns]].copy()

#   Hard assertions
assert len(final_cases) == 18, \
    f"Expected 18 final cases, got {len(final_cases)}"

assert final_cases['case_id'].is_unique, \
    f"Duplicate case_id values: {final_cases['case_id'].duplicated().sum()}"

_cell_counts = final_cases.groupby(
    ['difficulty_tier', 'correct'], observed=True
).size()
assert (_cell_counts == N_PER_CELL).all(), \
    f"Not exactly {N_PER_CELL} cases per difficulty×correct cell:
 ↪\n{_cell_counts}"

_required = ['case_id', 'pred_prob', 'y_true', 'difficulty_tier',
             'difficulty_score', 'correct', 'block']
```

```
_bad = [f for f in _required
        if f not in final_cases.columns or final_cases[f].isna().any()]
assert not _bad, f"Missing or null required fields: {_bad}"

assert (final_cases['purpose'] != 'other').all(), \
    "Selected case has purpose == 'other'"

assert final_cases['pred_prob'].between(0.05, 0.95, inclusive='neither').all(),␣
 ↪\
    (f"pred_prob outside (0.05, 0.95): "
     f"min={final_cases['pred_prob'].min():.4f}, "
     f"max={final_cases['pred_prob'].max():.4f}")

assert (final_cases['dti'] <= 60).all(), \
    f"DTI out of reasonable display range: max={final_cases['dti'].max():.1f}"

print("  All hard assertions passed")

#   Save
final_cases.to_csv(BUILD_DIR / "final_cases.csv", index=False)
protocol_rotation.to_csv(BUILD_DIR / "protocol_rotation.csv", index=False)

print("\nSaved to artifacts/build/:")
print(f"  final_cases.csv        ({len(final_cases)} cases)")
print(f"  protocol_rotation.csv")
print(f"\nColumns: {list(final_cases.columns)}")
print(f"Trials per participant : {len(final_cases)}")
```

```
  All hard assertions passed

Saved to artifacts/build/:
  final_cases.csv        (18 cases)
  protocol_rotation.csv

Columns: ['case_id', 'case_position', 'block', 'loan_amnt', 'term', 'int_rate',
'log_annual_inc', 'dti', 'revol_util', 'home_ownership', 'purpose',
'credit_history_years', 'pred_prob', 'y_true', 'difficulty_tier',
'difficulty_score', 'correct', 'confidence', 'conf_bin', 'model_optimal']
Trials per participant : 18
```

### Practice trials

[18]:
```
_selected_ids = set(cases['case_id'].tolist())

# Practice trial 1: clear non-default (very low pred_prob, y_true=0)
practice_safe = df_pool[
    ~df_pool['case_id'].isin(_selected_ids) &
    (df_pool['y_true'] == 0) &
```

```
    (df_pool['pred_prob'] < 0.06) &
    (df_pool['difficulty_tier'] == 'easy')
].sample(1, random_state=42)

# Practice trial 2: clear default (high pred_prob, y_true=1)
practice_risky = df_pool[
    ~df_pool['case_id'].isin(_selected_ids) &
    (df_pool['y_true'] == 1) &
    (df_pool['pred_prob'] > 0.55) &
    (df_pool['difficulty_tier'] == 'hard')
].sample(1, random_state=42)

practice_cases = pd.concat([practice_safe, practice_risky]).
 ↪reset_index(drop=True)
practice_cases['case_position'] = [-2, -1]
practice_cases['block'] = 'practice'

# Hard assertions
assert len(practice_cases) == 2, \
    f"Expected exactly 2 practice cases, got {len(practice_cases)}"
_overlap = set(practice_cases['case_id']) & _selected_ids
assert not _overlap, f"Practice cases overlap with final cases: {_overlap}"

print("=== PRACTICE TRIALS ===")
print(practice_cases[['case_id', 'pred_prob', 'y_true', 'difficulty_tier',
                      'purpose', 'loan_amnt', 'int_rate']].
 ↪to_string(index=False))

practice_cases.to_csv(BUILD_DIR / "practice_cases.csv", index=False)
print(f"\nSaved: {BUILD_DIR / 'practice_cases.csv'}")
print("  Practice case assertions passed")
```

```
=== PRACTICE TRIALS ===
 case_id  pred_prob  y_true difficulty_tier        purpose  loan_amnt  int_rate
  946400   0.059221       0            easy    credit_card       6600      8.39
  934189   0.623978       1            hard small_business      15000     24.99

Saved: /Users/annaasatryan/Desktop/capstone/artifacts/build/practice_cases.csv
  Practice case assertions passed
```

[19]:
```
selection_manifest = {
    "seed": 42,
    "input_candidate_pool": str(BUILD_DIR / "candidate_pool_scored.parquet"),
    "n_final_cases": int(len(final_cases)),
    "n_practice_cases": int(len(practice_cases)),
    "N_PER_CELL": int(N_PER_CELL),
    "probability_filter": {"min_exclusive": 0.05, "max_exclusive": 0.95},
```

```
    "excluded_purpose": "other",
    "cost_ratio": {
        "C_FN": int(C_FN),
        "C_FP": int(C_FP),
        "tau": round(float(tau), 6),
    },
    "sampling_cell_order": [list(c) for c in cells],
    "output_paths": {
        "final_cases": str(BUILD_DIR / "final_cases.csv"),
        "practice_cases": str(BUILD_DIR / "practice_cases.csv"),
        "protocol_rotation": str(BUILD_DIR / "protocol_rotation.csv"),
    },
}

with open(BUILD_DIR / "selection_manifest.json", "w") as f:
    json.dump(selection_manifest, f, indent=2)

print(f"Saved: {BUILD_DIR / 'selection_manifest.json'}")
```

```
Saved:
/Users/annaasatryan/Desktop/capstone/artifacts/build/selection_manifest.json
```

**Feature interpretation helpers**

[20]:
```
def dti_label(dti):
    if dti < 15: return "LOW"
    elif dti < 30: return "MODERATE"
    else: return "HIGH"

def util_label(x):
    if x < 30: return "LOW"
    elif x < 70: return "MODERATE"
    else: return "HIGH"

def income_label(x):
    if x < 30000: return "LOW"
    elif x < 100000: return "MEDIUM"
    else: return "HIGH"
```

**Full case preview**

INSTRUCTIONS

You are acting as a loan officer.

If you approve a loan and the borrower defaults → LARGE LOSS If you reject a good loan → SMALL LOSS

Your goal is to make decisions that minimize total loss.

Your performance will be evaluated based on decision quality.

[21]:
```
def render_case(row, protocol='ai_first', show_index=True):
    income = int(np.exp(row['log_annual_inc']))
    hist = round(row['credit_history_years'], 1)
    pred_pct = int(round(row['pred_prob'] * 100))
    tier = str(row['difficulty_tier']).upper()

    # labels
    dti_l = dti_label(row['dti'])
    util_l = util_label(row['revol_util'])
    inc_l = income_label(income)

    header = ""
    if show_index:
        header = (
            f"\n{'='*60}\n"
            f"CASE {int(row['case_position']):02d} of 18 | Block:␣
 ↪{row['block']} | [{tier}]\n"
            f"{'='*60}"
        )

    profile = (
        f"\nLOAN APPLICATION\n"
        f"{'-'*44}\n"
        f"Loan amount       : ${row['loan_amnt']:>10,.0f}\n"
        f"Interest rate     : {row['int_rate']:>9.2f}%\n"
        f"Annual income     : ${income:>10,.0f} ({inc_l})\n"
        f"Debt-to-income    : {row['dti']:>9.1f} ({dti_l})\n"
        f"Revolving util.   : {row['revol_util']:>9.1f}% ({util_l})\n"
        f"Credit history    : {hist:>7.1f} years\n"
        f"Home ownership    : {str(row['home_ownership']):>12}\n"
        f"Loan purpose      : {str(row['purpose']):>12}\n"
        f"Term              : {str(row['term']):>12}\n"
        f"{'-'*44}"
    )

    # ------------------------------------------
    # PROTOCOL LOGIC
    # ------------------------------------------

    if protocol == 'no_ai':
        decision_block = (
            f"\n\nDECISION TASK:\n"
            f"Would you APPROVE or REJECT this loan?\n"
            f"[Approve] [Reject]\n\n"
            f"(Optional) What probability do you assign to default?\n"
            f"[ 0% |------------------| 100% ]\n\n"
            f"How confident are you? (0-100%)\n"
```

```
            f"[ 0% |------------------| 100% ]\n"
        )

    elif protocol == 'human_first':
        decision_block = (
            f"\n\nSTEP 1 - Initial decision:\n"
            f"Would you APPROVE or REJECT?\n"
            f"[Approve] [Reject]\n\n"
            f"(Optional) Initial probability:\n"
            f"[ 0% |------------------| 100% ]\n"
            f"[SUBMIT]\n\n"
            f"--- AI revealed ---\n\n"
            f"AI ASSESSMENT:\n"
            f"Default probability: {pred_pct}%\n\n"
            f"STEP 2 - Final decision:\n"
            f"[Approve] [Reject]\n\n"
            f"(Optional) Final probability:\n"
            f"[ 0% |------------------| 100% ]\n\n"
            f"Confidence (0-100%):\n"
            f"[ 0% |------------------| 100% ]\n"
        )

    else:  # ai_first
        decision_block = (
            f"\n\nAI ASSESSMENT:\n"
            f"Default probability: {pred_pct}%\n\n"
            f"DECISION TASK:\n"
            f"Would you APPROVE or REJECT?\n"
            f"[Approve] [Reject]\n\n"
            f"(Optional) Probability estimate:\n"
            f"[ 0% |------------------| 100% ]\n\n"
            f"Confidence (0-100%):\n"
            f"[ 0% |------------------| 100% ]\n"
        )

    footer = f"\n{'-'*44}\n[SUBMIT]\n{'='*60}\n"

    print(header + profile + decision_block + footer)
```

**Summary table**

[22]:
```
summary = final_cases[[
    'case_position', 'block', 'difficulty_tier',
    'correct', 'pred_prob', 'confidence',
    'y_true', 'model_optimal',
    'home_ownership', 'purpose', 'loan_amnt', 'int_rate'
]].copy()
```

```
summary.columns = [
    'pos', 'block', 'difficulty',
    'model_correct', 'ai_pred', 'ai_conf',
    'true_label', 'normative_align',
    'ownership', 'purpose', 'loan_amnt', 'int_rate'
]

summary['ai_pred'] = summary['ai_pred'].round(3)
summary['ai_conf'] = summary['ai_conf'].round(3)

print("=== ALL 18 EXPERIMENTAL CASES ===\n")
print(summary.to_string(index=False))

print("\n=== KEY STATISTICS ===")
print(f"Cases where model is correct : "
      f"{int(summary['model_correct'].sum())} / 18")
print(f"Cases aligned with optimal   : "
      f"{int(summary['normative_align'].sum())} / 18")
print(f"True default rate in cases   : "
      f"{summary['true_label'].mean():.3f}")
print(f"Mean AI prediction           : "
      f"{summary['ai_pred'].mean():.3f}")
print(f"Mean AI confidence           : "
      f"{summary['ai_conf'].mean():.3f}")
print(f"Prediction range             : "
      f"[{summary['ai_pred'].min():.3f}, {summary['ai_pred'].max():.3f}]")

print("\n=== DESIGN BALANCE CHECK ===")
print(pd.crosstab(
    summary['difficulty'],
    [summary['model_correct'], summary['normative_align']],
    rownames=['difficulty'],
    colnames=['model_correct', 'normative_align']
))
```

```
=== ALL 18 EXPERIMENTAL CASES ===

 pos   block difficulty  model_correct  ai_pred  ai_conf  true_label
normative_align ownership            purpose  loan_amnt  int_rate
   1 block_1       easy              0    0.333    0.667           1
0      RENT debt_consolidation      30000     12.88
   2 block_1       easy              1    0.055    0.945           0
1  MORTGAGE                car       7200      7.97
   3 block_1     medium              0    0.234    0.766           1
0       OWN            medical       2000     15.77
   4 block_1     medium              1    0.169    0.831           0
0  MORTGAGE debt_consolidation       1000     13.99
```

```
   5 block_1       hard              0    0.900    0.900           0
1      RENT debt_consolidation      13125     24.85
   6 block_1       hard              1    0.946    0.946           1
1       OWN        credit_card      18000     28.72
   7 block_2       easy              0    0.370    0.630           1
0      RENT     small_business      21000     13.67
   8 block_2       easy              1    0.075    0.925           0
1  MORTGAGE debt_consolidation      10000      9.80
   9 block_2     medium              0    0.267    0.733           1
0      RENT debt_consolidation       8750     12.62
  10 block_2     medium              1    0.189    0.811           0
0  MORTGAGE        credit_card      32000     13.99
  11 block_2       hard              0    0.831    0.831           0
1      RENT   home_improvement      25350     30.99
  12 block_2       hard              1    0.923    0.923           1
1      RENT debt_consolidation      13200     28.14
  13 block_3       easy              0    0.397    0.603           1
0      RENT   home_improvement      12000     12.99
  14 block_3       easy              1    0.102    0.898           0
1       OWN   home_improvement      12000      7.99
  15 block_3     medium              0    0.296    0.704           1
0  MORTGAGE        credit_card      15500     18.99
  16 block_3     medium              1    0.210    0.790           0
0  MORTGAGE            medical       1200     13.44
  17 block_3       hard              0    0.779    0.779           0
1      RENT        credit_card      18900     30.17
  18 block_3       hard              1    0.858    0.858           1
1  MORTGAGE     small_business      25000     28.72

=== KEY STATISTICS ===
Cases where model is correct : 9 / 18
Cases aligned with optimal   : 9 / 18
True default rate in cases   : 0.500
Mean AI prediction           : 0.441
Mean AI confidence           : 0.808
Prediction range             : [0.055, 0.946]

=== DESIGN BALANCE CHECK ===
model_correct    0     1
normative_align  0  1  0  1
difficulty
easy             3  0  0  3
medium           3  0  3  0
hard             0  3  0  3
```